

# SALES DATA ANALYSIS & PREDICTION

End-to-End Data Analytics Capstone Project

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## PROJECT SCOPE

Data cleaning • Exploratory analysis • SQL/Pandas • Regression • Excel • Power BI • Business insights

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## 1. Project Overview

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This project presents an end-to-end analysis of sales data with the aim of understanding business performance, identifying meaningful patterns, and developing a regression-based approach for Sales prediction. The workflow follows a practical data analytics lifecycle: raw data is inspected and prepared, exploratory analysis is performed, business questions are examined through Python/Pandas and SQL, results are organized in Excel, interactive reporting is developed in Power BI, and a machine learning model is evaluated.

The report therefore combines descriptive, diagnostic, and predictive perspectives. Descriptive analysis explains what happened in the available sales data; diagnostic analysis examines relationships and differences between business dimensions; and predictive modeling investigates whether the available variables can be used to estimate Sales.

### Tools and Technologies

- Python
- Pandas and NumPy
- SQL
- Microsoft Excel
- Power BI
- Machine Learning / Linear Regression

### Project Components

- Data Cleaning
- Exploratory Data Analysis (EDA)
- SQL and Pandas Analysis
- Correlation Analysis
- Regression Modeling
- Power BI Dashboard
- Business Insights and Recommendations

## 2. Project Objectives

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1. Clean and prepare the sales dataset for reliable analysis.
2. Perform Exploratory Data Analysis (EDA) to understand sales and business performance.
3. Analyze the dataset using Python/Pandas and SQL.
4. Identify patterns across countries, products, segments, time periods, and discount bands.
5. Examine relationships among important numerical business variables.
6. Build a regression model to investigate Sales prediction.
7. Evaluate the regression model using MAE, MSE, RMSE, and  $R^2$ .
8. Create an interactive Power BI dashboard containing key business metrics and visualizations.
9. Summarize major findings and translate the analysis into practical business recommendations.

## 3. Theoretical Background

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### 3.1 Data Analytics

Data analytics is the systematic process of collecting, preparing, examining, and interpreting data to support evidence-based decision-making. In a business environment, analytics converts raw operational records into information that can be used to understand performance, discover patterns, monitor changes, and support planning.

A typical analytics workflow moves through several connected stages: data collection, data quality assessment, data cleaning, exploratory analysis, statistical analysis, visualization, modeling, interpretation, and communication. The quality of later conclusions depends strongly on the quality and structure of the earlier stages.

### 3.2 Data Cleaning and Data Quality

Data cleaning is the process of detecting and correcting or appropriately handling inaccurate, incomplete, duplicated, inconsistent, or incorrectly formatted data. It is a foundational step because analytical calculations assume that the underlying records are sufficiently consistent to represent the business process being studied.

Common data-quality checks include missing-value inspection, duplicate detection, data-type validation, range checks, inconsistent text identification, and date-format standardization. Cleaning should be documented because every transformation can affect subsequent results.

### 3.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is the process of investigating a dataset using summary statistics and visualizations before drawing conclusions or building models. EDA helps an analyst understand distributions, variation, trends, unusual observations, relationships between variables, and differences across categories.

For sales data, EDA can answer questions such as: Which countries contribute the most Sales? Which products generate substantial Sales or Profit? How does performance vary across segments? Are there visible monthly patterns? How are discounts associated with Sales? These questions provide context for deeper analysis.

### 3.4 Descriptive and Comparative Analysis

Descriptive analysis summarizes historical or observed data using measures such as totals, averages, counts, and grouped comparisons. Comparative analysis extends this by examining how a metric differs across categories, such as country, product, segment, year, or discount band.

Sales and Profit should not automatically be treated as interchangeable measures. Sales represents the amount generated from selling activity, whereas Profit reflects the amount remaining after relevant costs represented in the dataset. Examining both provides a more complete view of business performance.

### 3.5 Correlation Analysis

Correlation measures the strength and direction of association between numerical variables. A correlation value generally ranges from -1 to +1. A positive value indicates that the variables tend to increase together, a negative value indicates an inverse relationship, and a value near zero indicates a weak linear association.

Correlation is useful for identifying variables that move together and for generating hypotheses for further investigation. However, correlation by itself does not establish causation. A strong correlation can arise because variables are mathematically related, influenced by another factor, or measured from the same underlying business process.

### 3.6 Linear Regression

Linear regression is a supervised learning and statistical modeling technique used to estimate a continuous target variable from one or more explanatory variables. In a simple form, the model estimates a linear relationship between an input and a target. Multiple linear regression extends this idea to several predictors.

In this project, Sales is the target variable. Categorical variables are converted into numerical indicator variables using One-Hot Encoding so that they can be used by the regression algorithm. The data is divided into training and testing portions so that model performance can be assessed on observations not used to fit the model.

### 3.7 Regression Evaluation Metrics

MAE (Mean Absolute Error) measures the average absolute difference between actual and predicted Sales. It is easy to interpret because it is expressed in the same units as the target.

MSE (Mean Squared Error) averages the squared prediction errors. Because larger errors receive greater weight after squaring, MSE is sensitive to large prediction mistakes.

RMSE (Root Mean Squared Error) is the square root of MSE. It is expressed in the same units as Sales and is commonly used as an overall measure of prediction error.

$R^2$  (coefficient of determination) indicates the proportion of variation in the target that is explained by the fitted regression model, subject to the assumptions and evaluation design used.

### 3.8 Business Intelligence and Dashboards

Business Intelligence (BI) refers to technologies and practices used to transform business data into accessible information for analysis and decision-making. A dashboard presents selected metrics and visualizations in a compact interface so that users can monitor performance and investigate patterns interactively.

Power BI supports this process by combining data, measures, visualizations, filters, and interactive reporting. In this project, the dashboard acts as the reporting layer that communicates the results of the underlying analysis to a business audience.

## 4. Dataset Overview

The project uses a sales dataset containing transactional business information. The dataset was used as the primary source for data cleaning, exploratory analysis, business analysis, regression modeling, and dashboard development.

| Field               | Description                             |
|---------------------|-----------------------------------------|
| Country             | Geographic market                       |
| Segment             | Customer or business segment            |
| Product             | Product being sold                      |
| Discount Band       | Discount category                       |
| Units Sold          | Quantity sold                           |
| Manufacturing Price | Manufacturing cost per unit             |
| Sale Price          | Selling price per unit                  |
| Gross Sales         | Sales before discounts                  |
| Discounts           | Discount amount                         |
| Sales               | Sales amount after applicable discounts |
| COGS                | Cost of goods sold                      |
| Profit              | Profit generated                        |
| Date                | Transaction date                        |
| Month               | Month of transaction                    |
| Year                | Year of transaction                     |

## 5. Data Cleaning & Preparation

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The dataset was inspected before analysis to identify missing values, duplicate records, incorrect data types, and inconsistent text formatting. The purpose of preparation was to create a consistent analytical dataset while retaining the business meaning of the original fields.

### Cleaning Operations

- Checked the dataset structure and dimensions.
- Checked missing values.
- Checked duplicate records.
- Converted the Date column to a datetime format.
- Cleaned text fields and removed unnecessary spaces.
- Handled missing Discount Band values by assigning them to the Unknown category.
- Created useful date-related fields such as Year, Month Number, and Month Name.
- Saved the cleaned dataset as sales\_data\_cleaned.csv.

The cleaned dataset was then used for subsequent analysis and visualization. This separation between raw and cleaned data also improves reproducibility because the original source can be retained while the transformed version is used for analysis.

## 6. Exploratory Data Analysis

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Exploratory Data Analysis was performed to understand the distribution and relationships within the sales dataset. The analysis examined overall Sales, Profit, Units Sold, Discounts, country-level performance, product-level performance, segment performance, time trends, discount categories, and relationships among numerical variables.

### Sales by Country

Country-level analysis compares total Sales across geographic markets. This helps identify differences in market contribution and provides a basis for understanding geographic sales performance.

### Sales by Product

Product-level analysis compares total Sales generated by different products. This helps identify products that contribute substantially to overall revenue and supports product-level business analysis.

### Sales by Segment

Segment analysis examines how different customer or business segments contribute to total Sales. Comparing segments can reveal differences in the scale of business activity and help monitor changes in the composition of sales.

### Monthly Sales Trend

Time-series analysis examines how Sales change across months and years. Looking at Sales over time helps identify periods of higher or lower activity and provides a foundation for monitoring business performance.

### Profit by Product

Profit was analyzed across products to understand the relationship between sales activity and profitability. This is important because a product with high Sales is not necessarily the product with the highest Profit.

## Sales by Discount Band

Sales were compared across discount bands to understand how different discount categories are associated with Sales performance. The Unknown category represents records where the Discount Band value was unavailable and was handled during data cleaning.

## 7. Correlation Analysis

A correlation matrix was generated for the numerical variables in the dataset. The analysis examined Units Sold, Manufacturing Price, Sale Price, Gross Sales, Discounts, Sales, COGS, and Profit.

The correlation matrix provides a compact overview of how numerical variables move in relation to one another. It can help identify strong associations and variables that may be useful for further analysis and modeling. The results should be interpreted as associations rather than proof that one variable causes another.

## 8. Regression Model

### 8.1 Objective

A Linear Regression model was developed to investigate whether the available business variables could be used to predict Sales. Regression provides a quantitative framework for estimating a continuous outcome from explanatory variables.

### 8.2 Model Preparation

The dataset was divided into training and testing sets. Categorical variables were converted into numerical representations using One-Hot Encoding, while numerical variables were retained as numerical inputs. The model was trained using the training dataset and evaluated using the test dataset.

### 8.3 Model Evaluation

| Metric               | Result                | Meaning                                        |
|----------------------|-----------------------|------------------------------------------------|
| MAE                  | 5.235311359034052e-10 | Average absolute prediction error              |
| MSE                  | 4.760346197590459e-19 | Average squared prediction error               |
| RMSE                 | 6.899526213871833e-10 | Square root of MSE                             |
| R <sup>2</sup> Score | 1.0                   | Proportion of variation explained by the model |

The reported test metrics are extremely close to zero for MAE, MSE, and RMSE, while the R<sup>2</sup> score is 1.0. These values indicate an almost exact match between the model predictions and the observed test Sales values in this project. Because Sales is also closely related to other monetary fields in the dataset, such a result should be interpreted carefully: very high performance can occur when predictors contain information that is directly or mathematically linked to the target. The model should therefore be treated as an analytical result of this dataset and feature design rather than assumed to represent equally strong performance on unseen real-world sales.

### 8.4 Actual vs Predicted Sales

The Actual vs Predicted Sales visualization compares observed Sales values in the test dataset with the values predicted by the regression model. When points lie close to the ideal agreement line, predicted values are close to actual values. This visualization provides a graphical assessment that complements the numerical evaluation metrics.

## 9. Power BI Dashboard

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An interactive Power BI dashboard was created to present major business metrics and analytical results in a format suitable for monitoring and exploration.

### Key Performance Indicators

- Total Sales
- Total Profit
- Total Units Sold
- Total Discounts

### Dashboard Visualizations

- Sales by Country
- Sales by Product
- Sales by Segment
- Monthly Sales Trend
- Profit by Product
- Sales by Discount Band

### Interactive Filters

- Country
- Product
- Segment
- Year
- Discount Band

The dashboard combines high-level KPIs with categorical and time-based visualizations. This allows users to move from an overall performance view to more detailed slices of the dataset through filters.

## 10. Key Findings

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1. Sales performance varies across countries, indicating differences in geographic contribution.
2. Different products contribute different amounts to total Sales and Profit.
3. Sales contribution varies across customer/business segments.
4. Monthly sales analysis provides visibility into changes in Sales over time.
5. Product-level Profit analysis helps distinguish sales contribution from profitability.
6. Discount Band analysis provides information about Sales performance across different discount categories.
7. Correlation analysis provides insight into relationships among numerical business variables.
8. Regression analysis provides a quantitative approach for investigating Sales prediction.

## 11. Business Recommendations

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The recommendations below are framed as analytical actions supported by the dimensions examined in the project. They are intended to guide monitoring and further investigation rather than replace business judgment.

1. Monitor country-level performance regularly to identify markets with changing Sales contribution.
2. Review product-level Sales and Profit together rather than evaluating Sales alone.
3. Track segment performance to understand changes in customer/business-segment contribution.
4. Use monthly Sales trends for planning and monitoring business performance.
5. Evaluate discount levels together with Profit to understand their relationship with profitability.
6. Monitor products with lower profitability and investigate the factors contributing to their performance.

7. Use the Power BI dashboard for interactive monitoring of Sales, Profit, Units Sold, and Discounts.
8. Use the regression model as an analytical support tool for understanding Sales prediction, while considering the feature relationships and limitations of the current model.

## 12. Excel Analysis

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An Excel workbook was prepared containing the cleaned data and summarized analysis tables. Excel provides a practical tabular environment for reviewing results, checking calculations, and presenting summarized business information.

- Cleaned Data
- KPI Summary
- Country Analysis
- Product Analysis
- Segment Analysis
- Year Analysis
- Monthly Analysis
- Regression Model Metrics
- Actual vs Predicted Results

## 13. SQL Analysis

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SQL was used to prepare structured analytical queries for common business questions. SQL is particularly useful for filtering, grouping, aggregating, sorting, and summarizing records in a structured dataset or database environment.

- Total Sales, Profit, and Units Sold
- Sales by Country
- Profit by Product
- Sales by Segment
- Yearly Sales and Profit
- Monthly Sales
- Discount Band Analysis
- Sales and Profit by Country
- Top Sales Transactions
- Negative Profit Transactions

The SQL queries are stored in the sql folder as analysis\_queries.sql.

## 14. Project Deliverables

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1. Cleaned sales dataset
2. Python analysis notebook
3. SQL analysis queries
4. Excel analysis workbook
5. Power BI dashboard
6. Regression model results
7. Visualization screenshots
8. Final project report
9. README documentation

## 15. Project Structure

The project is organized into separate folders so that data, notebooks, SQL scripts, Excel files, Power BI assets, documentation, and screenshots can be located and maintained independently.

| Path / Folder Name | File Stored in Folder         | Purpose                           |
|--------------------|-------------------------------|-----------------------------------|
| Capstone-Project/  |                               | Main project folder               |
| data/              | sales_data.csv                | Original/raw sales dataset        |
|                    | sales_data_cleaned.csv        | Cleaned dataset used for analysis |
| notebooks/         | sales_analysis.ipynb          | Python analysis notebook          |
| sql/               | analysis_queries.sql          | SQL Business analysis queries     |
| excel/             | sales_analysis.xlsx           | Excel analysis workbook           |
|                    | regression_results.xlsx       | Regression results                |
| powerbi/           | sales_powerbi_data.csv        | Power BI dashboard data           |
|                    | Sales_Analysis_Dashboard.pbix | Power BI dashboard                |
| report/            | Final_Report.docx             | Final project report              |
| screenshots/       | sales_by_country.png          | Country sales visualization       |
|                    | sales_by_product.png          | Product sales visualization       |
|                    | sales_by_segment.png          | Segment sales visualization       |
|                    | sales_trend.png               | Sales trend visualization         |
|                    | correlation_heatmap.png       | Correlation visualization         |
|                    | actual_vs_predicted.png       | Regression visualization          |

## 16. Conclusion

This project demonstrates an end-to-end data analytics workflow, beginning with data cleaning and exploratory analysis and continuing through business analysis, correlation analysis, regression modeling, and interactive visualization. The workflow illustrates how the same dataset can support multiple analytical perspectives: Python and Pandas support preparation and exploration, SQL supports structured business questions, Excel supports organized tabular analysis, and Power BI supports interactive reporting.

The regression model provides a quantitative approach to Sales prediction, while the exploratory and dashboard analyses provide a broader view of Sales, Profit, Units Sold, Discounts, products, countries, segments, and time-based performance. The project also demonstrates an important analytical principle: model results must be interpreted in the context of the variables used and the relationships already present in the dataset.

Overall, the completed work documents the progression from raw sales records to a cleaned analytical dataset, business-focused findings, predictive analysis, and a presentation-ready reporting layer. This provides a structured foundation for further analysis, model refinement, and future business questions.